Jacob Kinyua

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Education

New York University **Sep 2023 – May 2027**

B.S in Computer Science, Minor in Cybersecurity

- **Relevant CourseWork:** Data Structures and Algorithms, Object-Oriented Programming, Design and Analysis of Algorithms, Computer Architecture and Organisation, Introduction to Operating Systems, Computer Networking, Computer Security, Application Security, Network Security, Software Engineering.
- **Courses currently taking:** Penetration Testing and Vulnerability Analysis, Databases, Design Project.

Technical Skills

Programming Languages: Python, C++, C, JavaScript, HTML, CSS, Ruby, SQL.
Framework/Tools: Next.js, Git, Linux, Ruby on Rails, GitHub, React, FastAPI,Docker, Bootstrap, Flask, BeautifulSoup, Turtle Graphics, Selenium.

Experience

Extern **Remote**

CyberSecurity Extern *June 2025 - Aug 2025*

- Built and secured an MQTT based IoT pipeline using, python, TLS for encryption and device certificates to simulate and protect real-time water monitoring systems.
- Conducted security testing, including replay simulations and TLS performance benchmarking, identifying vulnerabilities and tuning defenses.
- Achieved 100% protection against eavesdropping attempts, man-in-the middle attacks and replay attacks confirmed with benchmark results.

Lusoi Ranch and Resort **Nanyuki, Kenya**

Website Developer *June 2025 - Sep 2025*

- Utilised Javascript, HTML and CSS to revamp the hotel website optimizing API calls to improve booking speed and overall usability.
- Collaborated with the marketing department to highlight unique hotel amenities, seasonal offers, and featured packages, improving discoverability and booking flow.
- Achieved a 30% increase in web traffic and boosted revenue from online hotel package bookings by 10%.

Projects

Daily Motiv Inspiration Emailer — FastAPI, React, PostgreSQL, Gemini API **June 2026 – Sep 2026**

- Built a full-stack career-inspiration platform using **FastAPI**, **React**, and **PostgreSQL** to generate personalized role-model recommendations based on users' occupations, goals, career stages, and interests.
- Developed an AI-powered recommendation pipeline using the **Gemini API** to generate tailored insights, maintain recommendation history, and prevent repeated recommendations.
- Implemented passwordless email authentication, profile management, automated email delivery, and a React dashboard for viewing and managing recommendations.

Amazon Price Tracker — Python, Flask, PostgreSQL, Scrapingdog, Resend **June 2025 – Sep 2025**

- Developed a **Flask**-based Amazon price tracker with **PostgreSQL** to store product tracking requests, target prices, and price-check history.
- Integrated the **Scrapingdog API** to retrieve Amazon product data and implemented scheduled monitoring with **Render Cron Jobs** to check prices against user-defined targets.
- Integrated **Resend** for automated email alerts when tracked products reached target prices, with database-backed alert status to prevent duplicate notifications.

Weather Recommender — HTML, CSS, JavaScript **June 2025 – Sep 2025**

- Built a weather application using **HTML**, **CSS**, and **JavaScript** to display temperature, wind speed, weather descriptions, and icons for user-specified locations.
- Implemented a weather-based recommendation feature that suggests appropriate clothing and items to carry based on retrieved weather conditions.
- Developed interactive location search and reset functionality, alongside dedicated About and Feedback pages with like and comment features.

Extracurricular Activities

National Society for Black Engineers – Member — Brooklyn, NY Jan 2024 - Present
NYU OSIRIS LAB – Member — Brooklyn, NY Sep 2024 - Present